Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University of Alicante
Short name / acronym used in projects	NEUROVIS
Participant Identification Code (PIC), if available	999895886
Website(s)	https://www.ua.es
Country	Spain
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Carretera San Vicente del Raspeig s/n, Alicante, Spain.
Department(s), institute(s), research group(s) or unit(s) that would be involved	Please name the unit(s) most relevant to the potential MSCA-SE action
Additional institutional resources relevant to collaboration	Facilities and equipment's of the University of Alicante https://sstti.ua.es/ https://sstti.ua.es/es/infraestructuras-de-apoyo/animalario.html

Main contact person for this profile	Nicolás Cuenca. Full Professor of Cell Biology. Department of Physiology, Genetics and Microbiology
Main contact email	cuenca@ua.es

Other contact person(s), if relevant	Pedro Lax Zapata. Full Professor of Physiology. Department of Physiology, Genetics and Microbiology. pedro.lax@ua.es

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	X Retina
X Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
X Advanced therapies / regenerative medicine	X Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	X Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	X Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Retinal diseases, animal models, neuroprotection, gene therapy, prime editing, retinal organoids, cell culture, optomotor analysis, ERGs, Multielectrode arrays,
Geographies of interest for exchanges (optional)	Europe

Overview of your research / innovation focus
The NeuroVis Research Group at the University of Alicante investigates the cellular and molecular mechanisms underlying retinal degeneration and develops innovative therapeutic strategies to preserve or restore vision. Our research spans the entire translational pipeline, from fundamental studies of retinal circuitry and disease mechanisms to the preclinical evaluation of advanced therapies. A major focus of our work is the study of inherited retinal degenerations, glaucoma, diabetic retinopathy and retinal changes associated with neurodegenerative disorders such as Parkinson's disease. Using a combination of animal models and human donor retinas, we investigate neuronal remodeling, synaptic alterations, neuroinflammation and glial responses to identify therapeutic targets. The group has extensive expertise in retinal neuroprotection, gene therapy using AAV vectors, CRISPR-based genome editing, stem-cell-based regenerative approaches, and innovative photoactive biomaterials for retinal stimulation and vision restoration. We combine advanced retinal imaging, quantitative morphology, immunohistochemistry, electrophysiology, behavioral analysis and molecular biology to assess therapeutic efficacy at structural and functional levels. An additional research focus is the biology of intrinsically photosensitive melanopsin retinal ganglion cells (ipRGCs), investigating their role in circadian regulation, non-image-forming vision and neurodegenerative diseases, with particular emphasis on their potential as early biomarkers of neurological disorders. Through this multidisciplinary approach, NeuroVis aims to accelerate the translation of basic discoveries into clinically relevant therapies and diagnostic tools that improve visual function and patient quality of life.

Why your team could be relevant to a translational eye-health and vision consortium

The NeuroVis Research Group (University of Alicante) brings a multidisciplinary perspective that combines retinal neuroscience, translational ophthalmology, neurodegeneration, advanced imaging, gene and cell therapies, and bioengineering-oriented therapeutic development. Our added value lies in bridging fundamental mechanisms of neuronal degeneration with the preclinical validation of innovative therapeutic strategies, providing expertise that complements both basic and clinically oriented partners.

A major strength of our team is its internationally recognized expertise in **retinal neuroprotection**. For more than three decades, we have investigated the cellular and molecular mechanisms responsible for neuronal death and retinal remodeling in inherited and acquired retinal diseases. Our work has identified early pathogenic events that precede irreversible degeneration and has contributed to the development of therapeutic approaches aimed at preserving retinal structure and visual function. This expertise provides an essential framework for evaluating emerging therapies and identifying compounds of treatment efficacy.

Our research uniquely integrates **animal models with human pathological tissue**, allowing direct translation of experimental findings into clinically relevant contexts. We have extensive experience working with numerous models of inherited retinal degeneration, glaucoma and retinal injury, together with post-mortem human donor retinas affected by retinal diseases as well as systemic neurodegenerative disorders such as Parkinson's disease and diabetic retinopathy. This comparative approach enables us to identify conserved disease mechanisms, validate therapeutic targets, and discover retinal compounds with potential clinical application for both ocular and neurological diseases.

Another distinctive contribution is our expertise in the biology of **intrinsically photosensitive melanopsin retinal ganglion cells (ipRGCs)**. Our work has significantly advanced the understanding of these specialized neurons and their role in circadian photoentrainment, sleep regulation, cognitive function and non-image-forming vision. Because dysfunction of ipRGCs has emerged as an early indicator of several neurodegenerative disorders, this research opens new opportunities for biomarker discovery, patient stratification and translational studies extending beyond ophthalmology into neuroscience, ageing and precision medicine.

Our team also contributes broad experience in the development and preclinical assessment of **advanced therapeutic technologies**. These include AAV-mediated gene augmentation, CRISPR-based genome editing, stem-cell-based retinal replacement strategies and neuroprotective pharmacological interventions to restore vision after photoreceptor degeneration. We combine high-resolution retinal imaging, quantitative morphology, electrophysiology, behavioral analysis and molecular biology to provide comprehensive validation of therapeutic efficacy from cellular mechanisms to functional outcomes.

Importantly, our expertise extends beyond disease modelling. We offer advanced capabilities in retinal circuit analysis, synaptic connectivity, quantitative image analysis, histopathology and functional evaluation, providing enabling technologies that can support multiple work packages across a collaborative consortium. These complementary skills facilitate the validation of novel therapeutic approaches, medical devices, biomaterials and regenerative strategies developed by other partners.

The interdisciplinary nature of NeuroVis places us at the interface of neuroscience, ophthalmology, regenerative medicine, gene therapy, biomaterials and translational medicine. We believe this broad scientific perspective can stimulate new collaborations across traditionally separated disciplines, helping to generate innovative concepts, identify shared disease mechanisms, and accelerate the translation of laboratory discoveries into effective clinical solutions that improve visual health and patient quality of life.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

NeuroVis is a **multidisciplinary research group** integrating expertise in neuroscience, ophthalmology, cell and molecular biology, gene therapy, regenerative medicine, bioengineering and translational vision research. Our laboratory is fully equipped to perform comprehensive structural, molecular and functional studies, providing an integrated platform for preclinical ophthalmic research. Our main strengths include:

- **Retinal neurodegeneration and neuroprotection**, with more than 30 years of experience investigating the cellular and molecular mechanisms underlying inherited and acquired retinal diseases.

- **Animal models** of inherited retinal dystrophies, glaucoma, diabetic retinopathy and retinal injury for mechanistic studies and preclinical therapeutic evaluation.
- **Human retinal pathology**, including extensive analysis of post-mortem donor retinas from patients with inherited retinal diseases, diabetic retinopathy, Parkinson's disease and other neurodegenerative disorders.
- **Advanced retinal morphology and immunocytochemistry**, including multicolor immunolabelling, high-resolution confocal microscopy, three-dimensional image reconstruction, quantitative morphometry and retinal circuit analysis.
- **Functional phenotyping**, including electroretinography (ERG), multi-electrode array (MEA) recordings and **optomotor behavioral testing** for comprehensive assessment of visual function.
- **Gene therapy and genome editing**, including AAV-mediated gene augmentation, CRISPR-based genome editing and preclinical validation of innovative therapeutic strategies.
- **Cell-based regenerative medicine**, including stem-cell-derived retinal models and retinal cell replacement approaches.
- **Melanopsin retinal ganglion cell (ipRGC) biology**, with internationally recognized expertise in circadian regulation and non-image-forming vision.
- **Integrated translational analyses**, combining structural, molecular, electrophysiological and behavioral data to evaluate disease progression and therapeutic efficacy.
- **Comprehensive research infrastructure**, with dedicated facilities and state-of-the-art equipment to perform all experimental procedures in-house, from retinal histology and immunocytochemistry to confocal microscopy, electrophysiology, behavioral testing and quantitative image analysis.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Facilities and equipment of the NeuroVis Research Group (University of Alicante)

- **Animal research facilities:** Fully equipped laboratories for mouse and rat models of inherited retinal degeneration, glaucoma, diabetic retinopathy and retinal injury, including facilities for behavioral vision testing and experimental therapeutic interventions.
- **Retinal histology and immunocytochemistry laboratory:** Complete infrastructure for tissue processing, cryosectioning, paraffin embedding, multicolor immunocytochemistry, histological staining and quantitative retinal morphometry.
- **Advanced microscopy platform:** Light and electron microscopy. High-resolution laser scanning confocal microscopy, fluorescence microscopy, stereomicroscopy and three-dimensional image reconstruction for detailed structural and cellular analysis of retinal tissues.
- **Electrophysiology platform:** Electroretinography (ERG) and multi-electrode array (MEA) recording systems for functional assessment of retinal activity and evaluation of therapeutic efficacy.
- **Behavioral vision platform:** Optomotor response system for quantitative assessment of visual acuity and contrast sensitivity in experimental animal models.
- **Molecular and cell biology laboratory:** Facilities for DNA/RNA extraction, PCR and RT-PCR, protein analysis, immunoblotting, cell culture, transfection, CRISPR-based genome editing and AAV vector-based gene therapy studies.
- **Preclinical therapeutic platforms:** Infrastructure for evaluating neuroprotective therapies, gene augmentation, genome editing, stem-cell-based regenerative strategies and photoactive biomaterials for retinal stimulation and vision restoration.
- **Human retinal pathology platform:** Access to and analysis of post-mortem human donor retinas from patients with inherited retinal diseases, diabetic retinopathy, Parkinson's disease and other neurodegenerative disorders, enabling direct translational validation of experimental findings.
- **Image analysis and quantitative morphometry:** Advanced software platforms for three-dimensional reconstruction, retinal cell quantification, morphometric analysis and neuronal connectivity studies.
- **Integrated multidisciplinary infrastructure:** NeuroVis possesses the experimental facilities and specialized equipment required to perform the complete workflow of translational retinal research in-house, from molecular and cellular studies to structural, electrophysiological and behavioral assessment of therapeutic interventions.

Training capacity and knowledge-transfer potential

The NeuroVis Research Group provides a highly interdisciplinary training environment that combines retinal neuroscience, translational ophthalmology, molecular biology, electrophysiology, regenerative medicine and preclinical therapeutic development. Our laboratory has a strong commitment to research training and international knowledge exchange, having successfully supervised **18 PhD theses** over the last years. These doctoral projects have involved researchers from diverse disciplines, including **biology, ophthalmology, neurology and veterinary medicine**, reflecting the multidisciplinary nature of our research. The successful training of ophthalmologists, neurologists and veterinarians highlights the **translational scientific trajectory** of our group, demonstrating our ability to bridge basic neuroscience with clinical and biomedical research while fostering collaboration across different disciplines. We have also trained numerous undergraduate, MSc, PhD and postdoctoral researchers from national and international institutions.

Incoming researchers would receive hands-on training in experimental models of inherited and acquired retinal diseases, including retinal degeneration, glaucoma, diabetic retinopathy and neurodegenerative disorders such as Parkinson's disease. They would acquire practical experience in advanced retinal histology, multicolor immunocytochemistry, confocal microscopy, quantitative retinal morphometry, neuronal circuit analysis and molecular biology techniques.

Training also includes functional assessment of retinal therapies using electroretinography (ERG), multi-electrode array (MEA) recordings and optomotor behavioral testing, providing comprehensive expertise in evaluating retinal structure and function. Researchers would also gain experience in AAV-mediated gene therapy, CRISPR-based genome editing, neuroprotective strategies, stem-cell-based approaches and the evaluation of innovative photoactive biomaterials for vision restoration.

A distinctive strength of NeuroVis is the integration of animal models with human donor retinal tissue, allowing trainees to validate experimental findings in clinically relevant samples and to understand the translational pathway from basic discoveries to patient-oriented applications.
Beyond technical skills, visitors would benefit from an international, collaborative and multidisciplinary research environment that promotes scientific exchange, experimental design, critical thinking and the development of collaborative projects, facilitating long-term knowledge transfer and the establishment of sustainable research networks in vision science and translational ophthalmology.

What your team would hope to gain from participating
Participation in this consortium would provide an outstanding opportunity to strengthen the translational impact of our research by integrating complementary expertise, emerging technologies and multidisciplinary approaches that extend beyond our current capabilities. While NeuroVis has extensive experience in retinal neurodegeneration, neuroprotection and preclinical therapeutic development, we are particularly interested in collaborating with partners working in advanced gene and RNA therapies, retinal organoids, biomaterials, medical devices, optogenetics, artificial intelligence, multimodal imaging and precision medicine. A major objective is to establish long-term international collaborations that evolve into sustainable scientific partnerships through joint research projects, coordinated European funding applications, shared infrastructures, co-supervision of students, and high-impact scientific publications. We believe that combining our expertise in retinal neuroscience and translational ophthalmology with complementary disciplines such as bioengineering, computational sciences, nanotechnology and regenerative medicine will create new opportunities for innovation and accelerate the translation of research into clinical practice. The consortium would also provide an exceptional framework for the international training and mobility of our researchers. PhD students, postdoctoral fellows, clinicians and technical staff would benefit from exposure to new experimental models, technologies and research environments, while developing interdisciplinary skills, international networks and collaborative experience that will enhance their scientific careers. For our senior researchers, participation would facilitate the incorporation of novel methodologies, analytical tools and enabling technologies into our research programs, particularly in areas where collaboration with spesialized partners can substantially enhance the quality and translational relevance of our work. At the same time, we are committed to sharing our expertise in retinal neurobiology, retinal circuitry, human retinal pathology and preclinical evaluation of innovative therapies, ensuring a genuine bidirectional exchange of knowledge. Overall, we expect this consortium to become a long-term strategic partnership that promotes bidirectional knowledge transfer, joint publications, coordinated European funding applications, industrial collaboration and the co-development of innovative technologies, ultimately accelerating the translation of research into tangible benefits for patients.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Nicolas Cuenca	Full Professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	cuenca@ua.es

Pedro Lax	Full Professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	Pedro.lax@ua.es
Xavier Sánchez-Sáez	Assistant professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	xsanchez@ua.es
Isabel Ortuño Lizarán	Assistant professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	isortliz@ua.es
Carla Sánchez Castillo	Assistant professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	carla.sanchez@ua.es
Santiago Milla Navarro	Assistant professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	s.milla@gcloud.ua.es
Joaquín Jimenez Candela	PhD student	Department of Physiology, Genetics and Microbiology	Researcher	Early-Stage	joaquin.jimenez@ua.es
Marina Pastor Mas	Technician	Department of Physiology, Genetics and Microbiology	Technician	Experienced	marina.pastor@ua.es
Sergio Fuster Picher	PhD student	Department of Physiology, Genetics and Microbiology	Researcher	Early-Stage	Sergio.fuster@ua.es
Ceren Sahin	PhD student	Department of Physiology, Genetics and Microbiology	Researcher	Early-Stage	ceren.sahin@ua.es

Main disciplines represented in your team	Cell biology, Physiology, Ophthalmology, Neuroscience, Immunology, Gene Therapy, Molecular Biology, Neuroprotection, Animal Models, Inmunohistochemistry Inherited retinal diseases.
Experience in EU / international / multi-partner collaborative projects	Our research group is an active participant in the **European Union's Marie Skłodowska-Curie Actions (MSCA) Doctoral Networks** under the **Horizon Europe** program, including the **RETORNA** project. We also participate in the European **OPTOREMODE** project, funded through the **European Research Area Network (ERA-NET) for Research on Rare Diseases**, and collaborate in research programs supported by **The Michael J. Fox Foundation for Parkinson's Research**, **Bayer**, and **Boehringer Ingelheim**.
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes ☐ No X Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
1. Cuenca N, Ortuño-Lizarán I, Sánchez-Sáez X, Kutsyr O, Albertos-Arranz H, Fernández-Sánchez L, Martínez-Gil N, Noailles A, López-Garrido Ja, López-Gálvez M, Lax P, Maneu V, Pinilla I. Interpretation of OCT and OCTA images from a histological approach: Clinical and experimental implications. Prog Retin Eye Res. 2020 Jan 3:100828. IF 20,198 doi: 10.1016/j.preteyeres.2019.100828 2. Lee JY, Martin-Bastida A, Murueta-Goyena A, Gabilondo I, Cuenca N, Paola Piccini P, Jeon B. Multimodal brain and retina imaging of dopaminergic degeneration in Parkinson's disease. Nature reviews Neurology. 2022 Apr;18(4):203-220. IF 42,937. doi: 10.1038/s41582-022-00618-9

3. Kutsyr O, Noailles A, Martínez-Gil N, Maestre-Carballa L, Martinez-Garcia M, Maneu V, Cuenca N, Lax P. Short-term high-fat feeding exacerbates degeneration in retinitis pigmentosa by promoting retinal oxidative stress and inflammation. Proc Natl Acad Sci U S A. 2021 Oct 26;118(43):e2100566118. IF 11.205. doi: 10.1073/pnas.2100566118
4. Ruiz-Pastor MJ, Sánchez-Sáez X, Kutsyr O, Albertos-Arranz H, Sánchez-Castillo C, Ortuño-Lizarán I, Martínez-Gil N, Vidal-Gil L, Méndez L, Sánchez-Martín M, Maneu V, Lax P, Cuenca N. Prph2 knock-in mice recapitulate human central areolar choroidal dystrophy retinal degeneration and exhibit aberrant synaptic remodeling and microglial activation. Cell Death Dis. 2023 Nov 1;14(11):711. IF 9.00 doi: 10.1038/s41419-023-06243-8
5.Albertos-Arranz H, Martínez-Gil N, Sánchez-Sáez X, Molina-Martín JC, Lax P, Cuenca N. Neuronal Degeneration and Glial Activation in the Absence of Vascular Changes in Human Retinas of Patients with Diabetes. Invest Ophthalmol Vis Sci. 2025 Mar 3;66(3):53. IF 4,7. doi: 10.1167/iovs.66.3.53.

Up to 5 relevant previous projects, grants, networks or collaborations

Selected Research Projects

International

- **RETORNA – Targeting RNA as a Therapeutic Strategy for Retinal Diseases** (Project No. **101073316**)
  Funding: European Union, **Marie Skłodowska-Curie Actions (MSCA) Doctoral Networks**, Horizon Europe (HORIZON-MSCA-2021-DN-01-01).
  Participant: Nicolás Cuenca (NeuroVis Research Group).
  Duration: 2022–2027.
  Total Project Budget: €2,568,427.20.
  Funding to NeuroVis: €503,942.40.
- **OPTOREMODE. Retinitis Pigmentosa Diagnosis and Therapy: Retinal Remodelling and Optogenetic Reactivation of the Degenerated Retina**
  Funding: European Research Area Network for Research on Rare Diseases (E-Rare ERA-NET, JTC 2013), European Community.
  Principal Investigator: Nicolás Cuenca
  Duration: January 2013 – December 2016.

National

- **Optogenetic Therapy for Neurodegenerative Diseases**
  Funding: Generalitat Valenciana – European Regional Development Fund (ERDF/FEDER), Grant IDIFEDER/2017/064.
  Principal Investigator: Nicolás Cuenca
  Duration: February 2017 – December 2017.
  Funding: €414,391.33.
- **Therapy for Hereditary Retinal Dystrophies Using Neuroprotective Agents and CRISPR: A Translational Study from Animal Models to Human Retinal Organoids** (PROMETEO/2021/024)
  Funding: Ministry of Innovation, Universities, Science and Digital Society (Generalitat Valenciana).
  Principal Investigator: Nicolás Cuenca
  Duration: January 2021 – December 2023.
  Funding: €353,865.63.
- **Therapeutic Approaches for PRPH2-Associated Retinal Disorders: Prime Editing and Gene Augmentation for Targeted Gene Therapy** (PID2024-156526OB-I00)
  Funding: Spanish Ministry of Science, Innovation and Universities.
  Principal Investigators: Pedro Lax and Nicolás Cuenca.
  Duration: September 2025 – August 2028.
  Funding: €237,500.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Patents

1. New immortalized cell line MU-PH1 (ACC3187) derived from adult mouse retinal Müller cells for the treatment of retinal neurodegenerative diseases
Inventors: V.E. Maneu Flores, V. Gómez Vicente, M.L. Gil Herrero, D. Gozalbo Flor, **N. Cuenca Navarro**
National patent application: ES201201274
International application: PCT/ES2013/070775
Patent publications: WO2014096479-A1; WO2014096479-A8; ES2478694-A1; ES2478694-B2
Priority country: Europe
Patent granted: 19 February 2015
Patent holders: University of Alicante and University of Valencia
Commercial exploitation: Applied Biological Materials Inc. (Canada)
This patent describes the generation of the immortalized Müller glial cell line **MU-PH1 (ACC3187)**, a valuable experimental platform for retinal research and the development of novel therapeutic strategies for retinal neurodegenerative diseases.

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2. Use of 3-(2-Isothiocyanatoethyl)-5-methoxy-1H-indole for the treatment of retinal degenerative diseases
Inventors: R. León Martínez, L. Campello Blasco, P. Michalska Dziama, A. Noailles Gil, O. Kutsyr, L. Fernández Sánchez, V. Maneu Flores, P. Lax Zapata, A. García García, **N. Cuenca Navarro**
Spanish patent application: P201830704
Priority country: Spain
Priority date: 13 July 2018
Patent holders: University of Alicante (70%), Fundación para la Investigación Biomédica del Hospital Universitario de La Princesa (20%), and Autonomous University of Madrid (10%)
Substantive examination successfully completed: 2 June 2020
This patent covers the therapeutic use of **3-(2-isothiocyanatoethyl)-5-methoxy-1H-indole** as a novel neuroprotective compound for the treatment of retinal degenerative diseases.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only X Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	None

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
The NeuroVis Research Group offers an integrated preclinical platform for translational retinal research, combining disease modelling, therapeutic evaluation and comprehensive structural and functional analyses. Our key capabilities include: • **Preclinical animal models of retinal neurodegeneration**, including inherited retinal dystrophies (retinitis pigmentosa), ocular hypertension/glaucoma and diabetic retinopathy, for studying disease mechanisms and therapeutic interventions.

- **Comprehensive evaluation of neuroprotective therapies**, using complementary behavioral, functional, structural and molecular approaches to assess treatment efficacy. Our analytical platforms include **optomotor behavioral testing, electroretinography (ERG), optical coherence tomography (OCT), immunohistochemistry, confocal microscopy, Western blotting, PCR/qPCR, transcriptomic analysis and flow cytometry**.
- **Ex vivo and in vitro experimental platforms**, including **organotypic retinal explant cultures** and **retinal cell culture systems**, enabling mechanistic studies, drug screening and validation of novel therapeutic strategies.
- **Gene therapy and genome editing**, with expertise in **AAV-mediated gene delivery**, **CRISPR-based genome editing**, and the preclinical evaluation of gene therapies for inherited retinal diseases.
- **Integrated translational expertise**, combining animal models, human retinal pathology and multimodal functional and molecular analyses to accelerate the development of therapies for retinal and neurodegenerative diseases.

Capabilities or expertise you would especially like to access through the consortium

We are particularly interested in connecting with partners that provide complementary expertise and enabling technologies capable of expanding the translational scope of our research. In particular, we would welcome collaborations with teams offering:

- **Human retinal organoids and iPSC-derived retinal models** for disease modelling, target validation and therapeutic testing.
- **Single-cell and spatial omics technologies**, including single-cell RNA sequencing (scRNA-seq), spatial transcriptomics and multi-omics approaches to characterize retinal cell populations and therapeutic responses.
- **Advanced behavioral and functional vision assessment**, including innovative visual function tests and digital phenotyping platforms for both preclinical and clinical studies.
- **Artificial intelligence, machine learning and computational biology** for multimodal data integration, image analysis, biomarker discovery and predictive modelling.
- **Advanced retinal imaging technologies**, including adaptive optics, high-resolution OCT, OCT angiography and multimodal clinical imaging.
- **Clinical cohorts and patient-oriented research infrastructures**, facilitating validation of experimental findings in well-characterized patient populations and supporting precision medicine approaches.
- **Regenerative medicine and advanced cell therapies**, including retinal tissue engineering, cell replacement strategies and biofabrication technologies.
- **Nanoparticle-based gene delivery systems**, including non-viral vectors and targeted nanocarriers for retinal gene therapy and nucleic acid delivery.
- **Advanced gene therapy platforms**, particularly partners with expertise in the design, engineering and GMP production of **AAV vectors** and other viral vectors for preclinical and clinical gene therapy applications.
- **Biomaterials, nanotechnology and innovative drug delivery systems** for retinal therapeutics and sustained intraocular delivery.
- **Industrial partners and regulatory expertise**, providing knowledge on technology transfer, regulatory pathways, intellectual property, GMP manufacturing and clinical translation of advanced therapies and medical devices.
- **Large international clinical and translational networks**, promoting long-term collaborations, staff exchanges, joint grant applications and multidisciplinary research initiatives beyond the field of ophthalmology.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organization / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organization. Submission of this form confirms that the contributor agrees to this internal use. Organizations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

Name: Nicolás Cuenca
Date completed: 10/7/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)